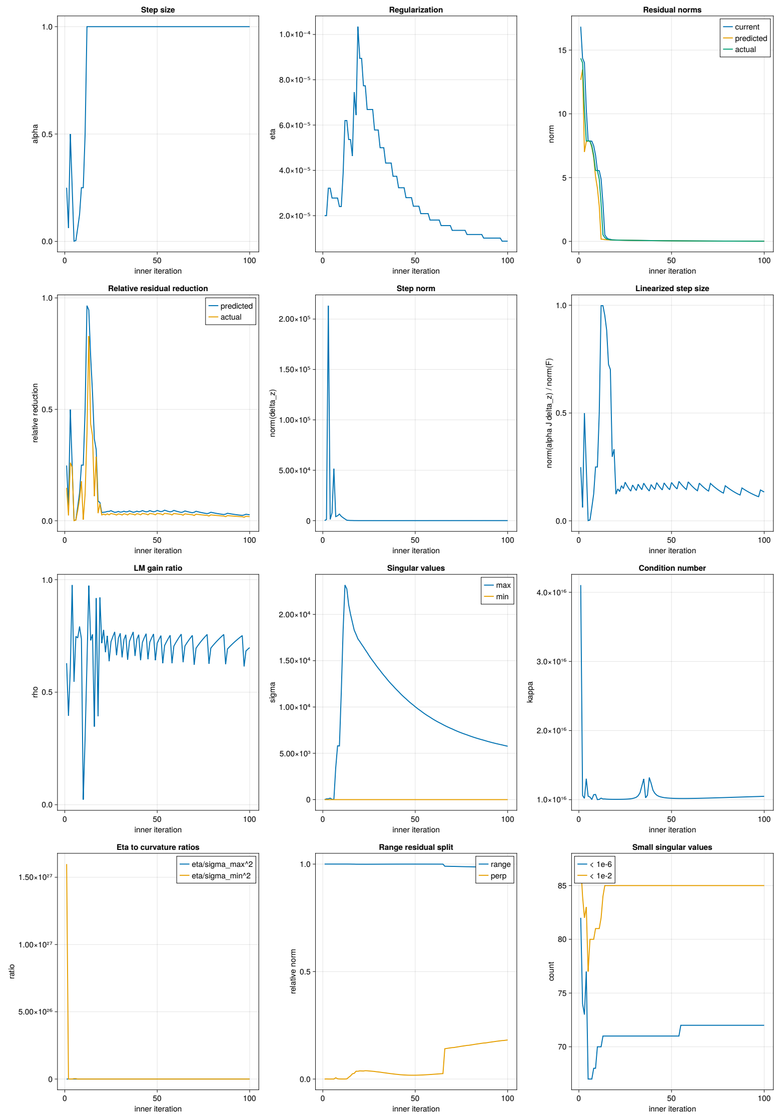
Solver Diagnostics

Recorded 100 inner iterations; reporting 100 rows (cap: 1000). Singular-value and range-space diagnostics use the raw ∇F SVD. If Marquardt column scaling is enabled for the step, the exact column space is unchanged, but numerical rank thresholding may differ.

Summary statistics

Metric	Minimum	Maximum	Mean	Median
`alpha`	9.765625e-04	1.000000e+00	9.125488e-01	1.000000e+00
`eta`	8.732233e-06	1.034460e-04	3.024953e-05	2.245856e-05
`kkt_error`	1.145437e-02	1.683624e+01	1.168830e+00	4.071881e-02
`norm_delta_z`	2.077520e+01	2.133356e+05	3.033378e+03	4.262338e+01
`norm_alpha_J_delta_z / norm_F`	9.757676e-04	9.980321e-01	2.029597e-01	1.522533e-01
`predicted_residual_norm`	1.114788e-02	1.345910e+01	8.664290e-01	3.890024e-02
`predicted_relative_reduction`	9.756499e-04	9.654218e-01	9.371447e-02	3.995134e-02
`actual_residual_norm`	1.124127e-02	1.435187e+01	1.000580e+00	3.940644e-02
`actual_relative_reduction`	5.330883e-04	8.288261e-01	5.824337e-02	2.852374e-02
`lm_delta_pred`	3.463732e-06	7.345429e+01	2.259809e+00	7.094418e-05
`lm_delta_actual`	2.290420e-06	4.442119e+01	1.417294e+00	4.871832e-05
`lm_gain_ratio`	2.281372e-02	9.759471e-01	6.999498e-01	7.244451e-01
`count(Jsvd.S < 1e-6)`	6.700000e+01	8.200000e+01	7.147000e+01	7.150000e+01
`count(Jsvd.S < 1e-2)`	7.700000e+01	8.700000e+01	8.457000e+01	8.500000e+01
`maximum(Jsvd.S)`	4.591747e+00	2.316496e+04	9.826486e+03	8.705501e+03
`minimum(Jsvd.S)`	1.118702e-16	2.266091e-12	9.487473e-13	8.552512e-13
`condition_number`	1.001169e+16	4.104531e+16	1.072834e+16	1.026620e+16
`eta_over_sigma_max_sq`	1.148854e-13	9.485802e-07	1.010517e-08	2.569177e-13
`eta_over_sigma_min_sq`	1.159119e+19	1.598090e+27	1.604910e+25	2.755076e+19
`range_rank`	3.960000e+02	4.110000e+02	4.006800e+02	4.000000e+02
`norm(F_range) / norm(F)`	9.833492e-01	1.000000e+00	9.951649e-01	9.995467e-01
`norm(F_perp) / norm(F)`	5.612347e-14	1.817262e-01	7.001995e-02	3.010247e-02
`residual_energy_svd_basis`	1.312027e-04	2.834590e+02	1.175318e+01	1.658484e-03
`residual_energy_frac_sigma_lt_1e_6`	3.150565e-27	1.626967e-03	3.541173e-04	9.529653e-05
`residual_energy_frac_sigma_lt_1e_4`	9.069972e-12	3.307087e-02	1.053036e-02	4.921899e-04
`residual_energy_frac_sigma_lt_1e_2`	7.456154e-05	9.166789e-01	7.405436e-01	8.716330e-01
`residual_energy_frac_sigma_ge_1e_2`	8.332110e-02	9.999254e-01	2.594564e-01	1.283670e-01
`weighted_inverse_sensitivity`	4.954243e+03	6.986880e+10	1.511933e+10	5.305775e+09
`residual_energy_sigma_at_50`	8.025729e-04	2.316496e+04	2.327865e+02	1.622641e-03
`residual_energy_sigma_at_90`	3.877299e-04	3.908207e+01	4.911199e-01	1.622641e-03
`residual_energy_sigma_at_99`	3.198494e-05	5.253708e+00	6.123418e-02	1.420851e-04
`maximum(abs.(delta_z[1:144]))`	9.243858e-04	4.420306e+00	1.906033e-01	4.790098e-03
`median(abs.(delta_z[1:144]))`	1.497407e-04	5.402795e-01	2.064207e-02	7.254541e-04
`maximum(abs.(delta_z[145:end]))`	1.220041e+01	2.066177e+05	2.817699e+03	4.206658e+01
`median(abs.(delta_z[145:end]))`	1.254924e-17	3.824504e-04	1.949543e-05	2.615976e-06

Metrics versus inner iteration



Per-iteration step diagnostics

Iter	α	η	$\|F\|$	$\|\delta z\|$	$\|\alpha J \delta z\|/\|F\|$	pred $\|F\|$	pred rel	actual $\|F\|$	actual rel	Δ_p	Δ_a	ρ
1	2.500000e-01	2.000000e-05	1.683624e+01	7.638804e+01	2.490550e-01	1.265938e+01	2.480873e-01	1.435187e+01	1.475608e-01	6.159951e+01	3.874139e+01	6.289236e-01
2	6.250000e-02	2.000000e-05	1.435187e+01	1.130932e+03	6.229707e-02	1.345910e+01	6.220591e-02	1.400582e+01	2.411177e-02	1.241442e+01	4.906577e+00	3.952320e-01
3	5.000000e-01	3.213061e-05	1.400582e+01	2.133356e+05	4.994664e-01	7.018156e+00	4.989116e-01	1.035957e+01	2.603383e-01	7.345429e+01	4.442119e+01	6.047461e-01
4	2.500000e-01	3.213061e-05	1.035957e+01	1.237804e+03	2.483257e-01	7.806687e+00	2.464275e-01	7.877808e+00	2.395624e-01	2.318817e+01	2.263043e+01	9.759471e-01
5	9.765625e-04	2.778221e-05	7.877808e+00	7.661221e+03	9.757676e-04	7.870122e+00	9.756499e-04	7.873608e+00	5.330883e-04	6.051915e-02	3.307456e-02	5.465140e-01
6	3.906250e-03	2.778221e-05	7.873608e+00	5.157579e+04	3.902681e-03	7.842883e+00	3.902216e-03	7.850638e+00	2.917312e-03	2.414408e-01	1.805912e-01	7.479729e-01
7	6.250000e-02	2.778221e-05	7.850638e+00	4.019490e+03	6.244642e-02	7.360453e+00	6.243896e-02	7.490377e+00	4.588938e-02	3.728130e+00	2.763384e+00	7.412253e-01
8	1.250000e-01	2.778221e-05	7.490377e+00	4.817072e+03	1.248979e-01	6.554962e+00	1.248823e-01	6.760623e+00	9.742560e-02	6.569112e+00	5.199866e+00	7.915629e-01
9	2.500000e-01	2.402229e-05	6.760623e+00	6.523398e+03	2.498275e-01	5.071838e+00	2.497973e-01	5.565860e+00	1.767237e-01	9.991241e+00	7.363609e+00	7.370065e-01
10	2.500000e-01	2.402229e-05	5.565860e+00	4.321632e+03	2.497055e-01	4.176267e+00	2.496637e-01	5.538047e+00	4.997232e-03	6.768797e+00	1.544215e-01	2.281372e-02
11	5.000000e-01	3.859255e-05	5.538047e+00	2.894766e+03	4.992694e-01	2.774339e+00	4.990401e-01	4.879773e+00	1.188639e-01	1.148650e+01	3.428889e+00	2.985147e-01
12	1.000000e+00	6.200012e-05	4.879773e+00	1.650351e+03	9.980321e-01	1.687337e-01	9.654218e-01	2.973293e+00	3.906903e-01	1.189186e+01	7.485855e+00	6.294943e-01
13	1.000000e+00	6.200012e-05	2.973293e+00	4.507949e+02	9.978700e-01	1.611548e-01	9.457992e-01	5.089501e-01	8.288261e-01	4.407250e+00	4.290720e+00	9.735596e-01
14	1.000000e+00	5.360932e-05	5.089501e-01	2.655164e+02	9.524789e-01	1.339513e-01	7.368086e-01	2.880710e-01	4.339897e-01	1.205436e-01	8.802265e-02	7.302141e-01
15	1.000000e+00	5.360932e-05	2.880710e-01	1.530282e+02	8.841992e-01	1.226161e-01	5.743544e-01	1.777073e-01	3.831127e-01	3.397508e-02	2.570250e-02	7.565102e-01
16	1.000000e+00	4.635408e-05	1.777073e-01	1.206101e+02	7.256210e-01	1.125308e-01	3.667634e-01	1.581537e-01	1.100328e-01	9.458357e-03	3.283651e-03	3.471692e-01
17	1.000000e+00	7.446926e-05	1.581537e-01	5.593070e+01	7.019459e-01	1.076961e-01	3.190415e-01	1.126885e-01	2.874749e-01	6.707070e-03	6.156950e-03	9.179791e-01
18	1.000000e+00	6.439094e-05	1.126885e-01	5.516186e+01	2.971362e-01	1.027069e-01	8.857645e-02	1.088753e-01	3.383778e-02	1.074989e-03	4.224255e-04	3.929578e-01
19	1.000000e+00	1.034460e-04	1.088753e-01	2.953549e+01	3.314597e-01	9.997443e-02	8.175326e-02	1.007090e-01	7.500630e-02	9.294770e-04	8.557682e-04	9.206987e-01
20	1.000000e+00	8.944612e-05	1.007090e-01	3.135806e+01	1.236516e-01	9.703049e-02	3.652617e-02	9.808115e-02	2.609354e-02	3.636938e-04	2.611958e-04	7.181749e-01
21	1.000000e+00	8.944612e-05	9.808115e-02	2.849161e+01	1.478388e-01	9.433318e-02	3.821294e-02	9.518248e-02	2.955386e-02	3.605815e-04	2.801044e-04	7.768129e-01
22	1.000000e+00	7.734091e-05	9.518248e-02	3.012942e+01	1.361988e-01	9.146462e-02	3.906025e-02	9.267707e-02	2.632215e-02	3.469631e-04	2.353324e-04	6.782634e-01
23	1.000000e+00	7.734091e-05	9.267707e-02	2.749746e+01	1.630221e-01	8.883443e-02	4.146266e-02	8.980980e-02	3.093831e-02	3.487415e-04	2.616198e-04	7.501825e-01
24	1.000000e+00	6.687395e-05	8.980980e-02	2.920564e+01	1.497835e-01	8.603683e-02	4.201068e-02	8.742110e-02	2.659729e-02	3.317320e-04	2.116755e-04	6.380918e-01
25	1.000000e+00	6.687395e-05	8.742110e-02	2.679151e+01	1.789966e-01	8.347164e-02	4.517740e-02	8.457886e-02	3.251206e-02	3.374668e-04	2.444326e-04	7.243158e-01
26	1.000000e+00	6.687395e-05	8.457886e-02	2.481665e+01	1.630406e-01	8.110133e-02	4.111579e-02	8.199577e-02	3.054061e-02	2.880786e-04	2.151386e-04	7.468051e-01
27	1.000000e+00	6.687395e-05	8.199577e-02	2.315223e+01	1.490758e-01	7.890056e-02	3.774846e-02	7.963087e-02	2.884176e-02	2.490043e-04	1.911156e-04	7.675195e-01
28	1.000000e+00	5.782355e-05	7.963087e-02	2.503546e+01	1.378515e-01	7.653780e-02	3.884260e-02	7.759055e-02	2.562212e-02	2.415203e-04	1.603904e-04	6.640865e-01
29	1.000000e+00	5.782355e-05	7.759055e-02	2.341060e+01	1.658942e-01	7.435153e-02	4.174508e-02	7.520648e-02	3.072634e-02	2.460720e-04	1.821397e-04	7.401885e-01
30	1.000000e+00	5.782355e-05	7.520648e-02	2.204632e+01	1.520947e-01	7.231586e-02	3.843582e-02	7.301552e-02	2.913264e-02	2.132157e-04	1.623745e-04	7.615505e-01
31	1.000000e+00	4.999798e-05	7.301552e-02	2.403489e+01	1.407797e-01	7.012480e-02	3.959043e-02	7.113602e-02	2.574110e-02	2.068890e-04	1.354664e-04	6.547781e-01
32	1.000000e+00	4.999798e-05	7.113602e-02	2.269376e+01	1.697139e-01	6.809251e-02	4.278434e-02	6.891649e-02	3.120125e-02	2.118715e-04	1.554255e-04	7.335838e-01
33	1.000000e+00	4.999798e-05	6.891649e-02	2.155386e+01	1.558938e-01	6.619645e-02	3.946857e-02	6.687258e-02	2.965764e-02	1.837560e-04	1.387697e-04	7.551845e-01
34	1.000000e+00	4.323149e-05	6.687258e-02	2.367548e+01	1.444940e-01	6.415216e-02	4.068073e-02	6.513240e-02	2.602237e-02	1.782215e-04	1.148564e-04	6.444588e-01
35	1.000000e+00	4.323149e-05	6.513240e-02	2.255920e+01	1.745345e-01	6.225304e-02	4.420785e-02	6.305476e-02	3.189872e-02	1.833945e-04	1.331634e-04	7.261038e-01
36	1.000000e+00	4.323149e-05	6.305476e-02	2.160142e+01	1.606776e-01	6.047907e-02	4.084853e-02	6.113903e-02	3.038195e-02	1.590927e-04	1.189607e-04	7.477446e-01
37	1.000000e+00	4.323149e-05	6.113903e-02	2.077520e+01	1.482775e-01	5.881662e-02	3.798584e-02	5.936520e-02	2.901308e-02	1.392936e-04	1.068771e-04	7.672796e-01
38	1.000000e+00	3.738074e-05	5.936520e-02	2.308940e+01	1.383115e-01	5.701696e-02	3.955585e-02	5.782294e-02	2.597928e-02	1.366467e-04	9.036761e-05	6.613231e-01
39	1.000000e+00	3.738074e-05	5.782294e-02	2.229507e+01	1.681082e-01	5.533867e-02	4.296332e-02	5.600652e-02	3.141344e-02	1.405617e-04	1.033809e-04	7.354840e-01
40	1.000000e+00	3.738074e-05	5.600652e-02	2.160605e+01	1.557205e-01	5.376601e-02	4.000445e-02	5.432233e-02	3.007136e-02	1.229732e-04	9.290750e-05	7.555100e-01
41	1.000000e+00	3.232181e-05	5.432233e-02	2.419661e+01	1.454379e-01	5.206405e-02	4.157180e-02	5.288331e-02	2.649039e-02	1.201249e-04	7.713550e-05	6.421273e-01
42	1.000000e+00	3.232181e-05	5.288331e-02	2.357132e+01	1.770757e-01	5.047759e-02	4.549105e-02	5.115827e-02	3.261969e-02	1.243286e-04	8.973780e-05	7.217795e-01
43	1.000000e+00	3.232181e-05	5.115827e-02	2.303005e+01	1.643473e-01	4.899183e-02	4.234781e-02	4.956017e-02	3.123839e-02	1.084846e-04	8.047918e-05	7.418487e-01
44	1.000000e+00	3.232181e-05	4.956017e-02	2.257805e+01	1.527961e-01	4.759670e-02	3.961784e-02	4.807545e-02	2.995794e-02	9.538215e-05	7.248081e-05	7.598991e-01
45	1.000000e+00	2.794753e-05	4.807545e-02	2.557896e+01	1.435034e-01	4.608419e-02	4.141941e-02	4.679726e-02	2.658714e-02	9.374800e-05	6.063261e-05	6.467617e-01
46	1.000000e+00	2.794753e-05	4.679726e-02	2.522061e+01	1.756741e-01	4.467203e-02	4.541344e-02	4.527096e-02	3.261517e-02	9.719640e-05	7.026189e-05	7.228857e-01
47	1.000000e+00	2.794753e-05	4.527096e-02	2.490663e+01	1.639166e-01	4.334789e-02	4.247915e-02	4.385300e-02	3.132168e-02	8.521021e-05	6.318720e-05	7.415450e-01
48	1.000000e+00	2.794753e-05	4.385300e-02	2.465712e+01	1.531519e-01	4.210334e-02	3.989823e-02	4.253277e-02	3.010566e-02	7.519705e-05	5.702425e-05	7.583310e-01
49	1.000000e+00	2.416524e-05	4.253277e-02	2.820020e+01	1.444951e-01	4.075312e-02	4.184182e-02	4.139873e-02	2.666273e-02	7.410982e-05	4.759084e-05	6.421665e-01
50	1.000000e+00	2.416524e-05	4.139873e-02	2.806582e+01	1.777038e-01	3.949178e-02	4.606296e-02	4.003889e-02	3.284735e-02	7.712701e-05	5.537102e-05	7.179200e-01
51	1.000000e+00	2.416524e-05	4.003889e-02	2.795341e+01	1.665524e-01	3.830869e-02	4.321301e-02	3.877398e-02	3.159216e-02	6.777855e-05	4.984580e-05	7.354215e-01
52	1.000000e+00	2.416524e-05	3.877398e-02	2.786017e+01	1.562717e-01	3.719659e-02	4.068158e-02	3.759519e-02	3.040156e-02	5.991749e-05	4.501159e-05	7.512261e-01
53	1.000000e+00	2.089483e-05	3.759519e-02	3.205744e+01	1.479785e-01	3.599015e-02	4.269255e-02	3.659399e-02	2.663117e-02	5.905351e-05	3.713925e-05	6.289084e-01
54	1.000000e+00	2.089483e-05	3.659399e-02	3.208416e+01	1.826952e-01	3.486337e-02	4.729225e-02	3.537880e-02	3.320720e-02	6.183248e-05	4.373008e-05	7.072348e-01
55	1.000000e+00	2.089483e-05	3.537880e-02	3.206793e+01	1.718670e-01	3.380688e-02	4.443109e-02	3.424817e-02	3.195802e-02	5.437714e-05	3.936140e-05	7.238592e-01
56	1.000000e+00	2.089483e-05	3.424817e-02	3.206421e+01	1.618117e-01	3.281427e-02	4.186782e-02	3.319466e-02	3.076105e-02	4.808028e-05	3.552582e-05	7.388856e-01
57	1.000000e+00	2.089483e-05	3.319466e-02	3.206804e+01	1.527595e-01	3.187989e-02	3.960770e-02	3.220984e-02	2.966782e-02	4.277884e-05	3.220560e-05	7.528395e-01
58	1.000000e+00	1.806702e-05	3.220984e-02	3.699295e+01	1.454766e-01	3.086499e-02	4.175290e-02	3.137113e-02	2.603895e-02	4.241323e-05	2.666302e-05	6.286486e-01
59	1.000000e+00	1.806702e-05	3.137113e-02	3.709083e+01	1.806617e-01	2.991600e-02	4.638448e-02	3.035310e-02	3.245124e-02	4.459048e-05	3.141862e-05	7.046038e-01
60	1.000000e+00	1.806702e-05	3.035310e-02	3.709692e+01	1.708508e-01	2.902541e-02	4.374133e-02	2.940377e-02	3.127638e-02	3.941798e-05	2.836464e-05	7.195863e-01
61	1.000000e+00	1.806702e-05	2.940377e-02	3.708831e+01	1.616562e-01	2.818808e-02	4.134451e-02	2.851750e-02	3.014106e-02	3.500675e-05	2.566667e-05	7.331920e-01
62	1.000000e+00	1.806702e-05	2.851750e-02	3.706435e+01	1.533062e-01	2.739943e-02	3.920660e-02	2.768781e-02	2.909421e-02	3.125964e-05	2.331662e-05	7.459015e-01
63	1.000000e+00	1.806702e-05	2.768781e-02	3.702397e+01	1.456946e-01	2.665541e-02	3.728714e-02	2.690913e-02	2.812369e-02	2.805195e-05	2.125686e-05	7.577678e-01
64	1.000000e+00	1.562192e-05	2.690913e-02	4.262446e+01	1.395980e-01	2.584557e-02	3.952400e-02	2.624040e-02	2.485134e-02	2.805380e-05	1.777128e-05	6.334716e-01
65	1.000000e+00	1.562192e-05	2.624040e-02	4.262229e+01	1.744957e-01	2.508676e-02	4.396435e-02	2.543228e-02	3.079680e-02	2.960658e-05	2.087887e-05	7.052103e-01
66	1.000000e+00	1.562192e-05	2.543228e-02	4.249574e+01	1.659428e-01	2.437337e-02	4.163636e-02	2.467624e-02	2.972743e-02	2.636979e-05	1.894193e-05	7.183194e-01
67	1.000000e+00	1.562192e-05	2.467624e-02	4.233910e+01	1.578481e-01	2.370157e-02	3.949852e-02	2.396834e-02	2.868767e-02	2.357632e-05	1.721784e-05	7.303023e-01
68	1.000000e+00	1.562192e-05	2.396834e-02	4.215549e+01	1.504226e-01	2.306791e-02	3.756722e-02	2.330395e-02	2.771959e-02	2.117628e-05	1.570368e-05	7.415691e-01
69	1.000000e+00	1.562192e-05	2.330395e-02	4.194694e+01	1.435900e-01	2.246935e-02	3.581355e-02	2.267906e-02	2.681471e-02	1.910113e-05	1.436713e-05	7.521612e-01
70	1.000000e+00	1.350772e-05	2.267906e-02	4.808300e+01	1.380989e-01	2.181701e-02	3.801085e-02	2.214662e-02	2.347715e-02	1.917892e-05	1.193348e-05	6.222187e-01
71	1.000000e+00	1.350772e-05	2.214662e-02	4.784942e+01	1.734325e-01	2.120502e-02	4.251640e-02	2.149616e-02	2.937071e-02	2.040983e-05	1.419398e-05	6.954481e-01
72	1.000000e+00	1.350772e-05	2.149616e-02	4.748300e+01	1.655507e-01	2.062906e-02	4.033749e-02	2.088653e-02	2.835981e-02	1.826341e-05	1.291881e-05	7.073606e-01
73	1.000000e+00	1.350772e-05	2.088653e-02	4.708739e+01	1.580417e-01	2.008613e-02	3.832146e-02	2.031478e-02	2.737397e-02	1.639730e-05	1.177837e-05	7.183112e-01
74	1.000000e+00	1.350772e-05	2.031478e-02	4.666769e+01	1.511030e-01	1.957357e-02	3.648615e-02	1.977743e-02	2.645144e-02	1.478279e-05	1.077188e-05	7.286771e-01
75	1.000000e+00	1.350772e-05	1.977743e-02	4.622757e+01	1.446752e-01	1.908901e-02	3.480842e-02	1.927141e-02	2.558554e-02	1.337823e-05	9.879670e-06	7.384884e-01
76	1.000000e+00	1.350772e-05	1.927141e-02	4.576904e+01	1.387064e-01	1.863027e-02	3.326882e-02	1.879407e-02	2.476935e-02	1.215009e-05	9.085096e-06	7.477390e-01
77	1.000000e+00	1.350772e-05	1.879407e-02	4.529603e+01	1.331606e-01	1.819544e-02	3.185216e-02	1.834298e-02	2.400185e-02	1.107155e-05	8.376122e-06	7.565448e-01
78	1.000000e+00	1.167965e-05	1.834298e-02	5.163974e+01	1.287095e-01	1.771985e-02	3.397065e-02	1.795505e-02	2.114841e-02	1.123579e-05	7.040452e-06	6.266096e-01
79	1.000000e+00	1.167965e-05	1.795505e-02	5.107719e+01	1.626688e-01	1.727208e-02	3.803765e-02	1.748261e-02	2.631258e-02	1.202950e-05	8.371150e-06	6.958848e-01
80	1.000000e+00	1.167965e-05	1.748261e-02	5.041069e+01	1.560368e-01	1.684925e-02	3.622788e-02	1.703785e-02	2.543996e-02	1.087218e-05	7.676605e-06	7.060780e-01
81	1.000000e+00	1.167965e-05	1.703785e-02	4.973422e+01	1.496697e-01	1.644940e-02	3.453765e-02	1.661893e-02	2.458776e-02	9.852745e-06	7.049795e-06	7.155158e-01
82	1.000000e+00	1.167965e-05	1.661893e-02	4.905217e+01	1.437335e-01	1.607079e-02	3.298304e-02	1.622365e-02	2.378520e-02	8.959316e-06	6.491083e-06	7.245065e-01
83	1.000000e+00	1.167965e-05	1.622365e-02	4.836768e+01	1.381886e-01	1.571181e-02	3.154877e-02	1.585005e-02	2.302783e-02	8.172858e-06	5.991290e-06	7.330717e-01

Iter	α	η	$\|F\|$	$\|\delta z\|$	$\|\alpha J\delta z\|/\|F\|$	pred $\|F\|$	pred rel	actual $\|F\|$	actual rel	Δ_p	Δ_a	ρ
84	1.000000e+00	1.167965e-05	1.585005e-02	4.768330e+01	1.329996e-01	1.537103e-02	3.022173e-02	1.549641e-02	2.231174e-02	7.477698e-06	5.542715e-06	7.412328e-01
85	1.000000e+00	1.167965e-05	1.549641e-02	4.700118e+01	1.281353e-01	1.504716e-02	2.899060e-02	1.516117e-02	2.163353e-02	6.860852e-06	5.138854e-06	7.490110e-01
86	1.000000e+00	1.167965e-05	1.516117e-02	4.632310e+01	1.235677e-01	1.473899e-02	2.784559e-02	1.484293e-02	2.099019e-02	6.311500e-06	4.774187e-06	7.564268e-01
87	1.000000e+00	1.009898e-05	1.484293e-02	5.260493e+01	1.198992e-01	1.440062e-02	2.979956e-02	1.456840e-02	1.849594e-02	6.467397e-06	4.037203e-06	6.242393e-01
88	1.000000e+00	1.009898e-05	1.456840e-02	5.179910e+01	1.524120e-01	1.408074e-02	3.347328e-02	1.423274e-02	2.304010e-02	6.985406e-06	4.833655e-06	6.919649e-01
89	1.000000e+00	1.009898e-05	1.423274e-02	5.093017e+01	1.468100e-01	1.377752e-02	3.198369e-02	1.391530e-02	2.230361e-02	6.375352e-06	4.467676e-06	7.007733e-01
90	1.000000e+00	1.009898e-05	1.391530e-02	5.007287e+01	1.413988e-01	1.348973e-02	3.058297e-02	1.361495e-02	2.158407e-02	5.831394e-06	4.134338e-06	7.089793e-01
91	1.000000e+00	1.009898e-05	1.361495e-02	4.922950e+01	1.363176e-01	1.321624e-02	2.928489e-02	1.333034e-02	2.090381e-02	5.348961e-06	3.834374e-06	7.168445e-01
92	1.000000e+00	1.009898e-05	1.333034e-02	4.840144e+01	1.315394e-01	1.295604e-02	2.807916e-02	1.306027e-02	2.025974e-02	4.919560e-06	3.563649e-06	7.243836e-01
93	1.000000e+00	1.009898e-05	1.306027e-02	4.758968e+01	1.270399e-01	1.270821e-02	2.695676e-02	1.280365e-02	1.964910e-02	4.536062e-06	3.318634e-06	7.316113e-01
94	1.000000e+00	1.009898e-05	1.280365e-02	4.679490e+01	1.227969e-01	1.247191e-02	2.590979e-02	1.255949e-02	1.906939e-02	4.192457e-06	3.096306e-06	7.385421e-01
95	1.000000e+00	1.009898e-05	1.255949e-02	4.601756e+01	1.187907e-01	1.224637e-02	2.493128e-02	1.232691e-02	1.851841e-02	3.883659e-06	2.894063e-06	7.451899e-01
96	1.000000e+00	1.009898e-05	1.232691e-02	4.525791e+01	1.150033e-01	1.203088e-02	2.401506e-02	1.210510e-02	1.799413e-02	3.605338e-06	2.709658e-06	7.515683e-01
97	1.000000e+00	8.732233e-06	1.210510e-02	5.130584e+01	1.119324e-01	1.179322e-02	2.576457e-02	1.191439e-02	1.575480e-02	3.726736e-06	2.290420e-06	6.145913e-01
98	1.000000e+00	8.732233e-06	1.191439e-02	5.040635e+01	1.430611e-01	1.156754e-02	2.911181e-02	1.167853e-02	1.979616e-02	4.072345e-06	2.782303e-06	6.832188e-01
99	1.000000e+00	8.732233e-06	1.167853e-02	4.947685e+01	1.383208e-01	1.135268e-02	2.790186e-02	1.145437e-02	1.919362e-02	3.752389e-06	2.592657e-06	6.909350e-01
100	1.000000e+00	8.732233e-06	1.145437e-02	4.857229e+01	1.337168e-01	1.114788e-02	2.675784e-02	1.124127e-02	1.860421e-02	3.463732e-06	2.418216e-06	6.981535e-01

Per-iteration SVD and range diagnostics

Iter	$\sigma_{\max}$	$\sigma_{\min}$	κ	$\eta/\sigma^2_{\max}$	$\eta/\sigma^2_{\min}$	rank	$\|F_r\|/\|F\|$	$\|F_\perp\|/\|F\|$	$N_{<10^{-6}}$	$N_{<10^{-2}}$
1	4.591747e+00	1.118702e-16	4.104531e+16	9.485802e-07	1.598090e+27	396	1.000000e+00	5.612347e-14	82	87
2	7.924901e+01	7.469383e-15	1.060985e+16	3.184508e-09	3.584764e+23	404	9.999999e-01	3.925855e-04	74	84
3	8.386632e+01	8.227705e-15	1.019316e+16	4.568186e-09	4.746370e+23	405	1.000000e+00	8.772789e-05	73	82
4	1.588648e+02	1.219395e-14	1.302816e+16	1.273103e-09	2.160876e+23	401	1.000000e+00	3.011639e-06	77	83
5	3.243232e+01	3.086407e-15	1.050812e+16	2.641257e-08	2.916489e+24	411	1.000000e+00	9.469944e-05	67	77
6	3.239576e+01	3.121338e-15	1.037880e+16	2.647223e-08	2.851577e+24	411	1.000000e+00	9.505830e-05	67	80
7	3.439848e+03	3.430715e-13	1.002662e+16	2.347947e-12	2.360464e+20	408	9.999827e-01	5.874718e-03	67	80
8	5.805681e+03	5.413866e-13	1.072372e+16	8.242528e-13	9.478763e+19	408	9.999990e-01	1.448444e-03	68	80
9	5.791877e+03	5.384887e-13	1.075580e+16	7.161035e-13	8.284406e+19	408	9.999999e-01	3.868940e-04	68	81
10	1.157532e+04	1.156180e-12	1.001169e+16	1.792869e-13	1.797063e+19	405	1.000000e+00	1.962821e-04	70	81
11	1.832818e+04	1.824685e-12	1.004457e+16	1.148854e-13	1.159119e+19	405	1.000000e+00	1.355668e-04	70	81
12	2.316496e+04	2.266091e-12	1.022243e+16	1.155392e-13	1.207363e+19	405	1.000000e+00	1.452810e-04	70	82
13	2.275083e+04	2.251119e-12	1.010645e+16	1.197838e-13	1.223476e+19	404	9.999997e-01	8.324270e-04	71	84
14	2.103563e+04	2.082011e-12	1.010351e+16	1.211517e-13	1.236728e+19	404	9.999684e-01	7.951153e-03	71	85
15	2.001567e+04	1.985867e-12	1.007906e+16	1.338135e-13	1.359377e+19	403	9.998933e-01	1.460998e-02	71	85
16	1.915116e+04	1.902430e-12	1.006669e+16	1.263856e-13	1.280769e+19	403	9.997078e-01	2.417438e-02	71	85
17	1.830015e+04	1.819521e-12	1.005767e+16	2.223657e-13	2.249379e+19	402	9.996480e-01	2.653071e-02	71	85
18	1.784499e+04	1.774901e-12	1.005408e+16	2.022051e-13	2.043979e+19	402	9.993404e-01	3.631541e-02	71	85
19	1.735555e+04	1.726578e-12	1.005199e+16	3.434292e-13	3.470096e+19	402	9.993479e-01	3.610673e-02	71	85
20	1.707782e+04	1.699011e-12	1.005163e+16	3.066878e-13	3.098627e+19	402	9.992785e-01	3.797947e-02	71	85
21	1.677120e+04	1.668406e-12	1.005223e+16	3.180044e-13	3.213349e+19	402	9.992915e-01	3.763625e-02	71	85
22	1.648343e+04	1.639498e-12	1.005395e+16	2.846520e-13	2.877315e+19	402	9.993032e-01	3.732522e-02	71	85
23	1.617038e+04	1.607796e-12	1.005748e+16	2.957799e-13	2.991901e+19	401	9.992467e-01	3.880882e-02	71	85
24	1.587786e+04	1.577844e-12	1.006301e+16	2.652608e-13	2.686143e+19	401	9.992632e-01	3.837963e-02	71	85
25	1.556088e+04	1.544906e-12	1.007238e+16	2.761779e-13	2.801903e+19	401	9.992972e-01	3.748530e-02	71	85
26	1.526583e+04	1.513566e-12	1.008601e+16	2.869564e-13	2.919136e+19	401	9.993225e-01	3.680372e-02	71	85
27	1.498982e+04	1.483365e-12	1.010528e+16	2.976213e-13	3.039213e+19	401	9.993504e-01	3.603877e-02	71	85
28	1.473088e+04	1.453839e-12	1.013240e+16	2.664696e-13	2.735721e+19	401	9.993798e-01	3.521474e-02	71	85
29	1.444978e+04	1.419756e-12	1.017766e+16	2.769376e-13	2.868650e+19	401	9.994215e-01	3.400854e-02	71	85
30	1.418761e+04	1.384553e-12	1.024707e+16	2.872675e-13	3.016378e+19	401	9.994547e-01	3.301857e-02	71	85
31	1.394190e+04	1.346208e-12	1.035642e+16	2.572222e-13	2.758851e+19	401	9.994875e-01	3.201293e-02	71	85
32	1.367547e+04	1.293388e-12	1.057337e+16	2.673426e-13	2.988786e+19	401	9.995300e-01	3.065686e-02	71	85
33	1.342718e+04	1.220230e-12	1.100381e+16	2.773209e-13	3.357910e+19	400	9.995634e-01	2.954807e-02	71	85
34	1.319470e+04	1.100796e-12	1.198651e+16	2.483140e-13	3.567685e+19	400	9.995948e-01	2.846472e-02	71	85
35	1.294283e+04	9.955854e-13	1.300022e+16	2.580725e-13	4.361574e+19	400	9.996333e-01	2.707872e-02	71	85
36	1.270830e+04	1.236616e-12	1.027667e+16	2.676859e-13	2.827032e+19	400	9.996628e-01	2.596853e-02	71	85
37	1.248885e+04	1.174049e-12	1.063741e+16	2.771760e-13	3.136373e+19	400	9.996894e-01	2.492223e-02	71	85
38	1.228292e+04	9.309140e-13	1.319447e+16	2.477677e-13	4.313490e+19	400	9.997133e-01	2.394573e-02	71	85
39	1.205940e+04	9.757978e-13	1.235850e+16	2.570377e-13	3.925801e+19	400	9.997407e-01	2.277139e-02	71	85
40	1.185081e+04	1.042339e-12	1.136944e+16	2.661658e-13	3.440567e+19	400	9.997609e-01	2.186526e-02	71	85
41	1.165526e+04	1.066822e-12	1.092521e+16	2.379316e-13	2.839957e+19	400	9.997782e-01	2.105960e-02	71	85
42	1.144320e+04	1.073453e-12	1.066017e+16	2.468318e-13	2.804977e+19	400	9.997977e-01	2.011608e-02	71	85
43	1.124547e+04	1.070173e-12	1.050809e+16	2.555880e-13	2.822201e+19	400	9.998106e-01	1.946361e-02	71	85
44	1.106025e+04	1.062151e-12	1.041306e+16	2.642203e-13	2.864989e+19	400	9.998208e-01	1.893288e-02	71	85
45	1.088625e+04	1.051860e-12	1.034952e+16	2.358233e-13	2.525965e+19	400	9.998285e-01	1.851690e-02	71	85
46	1.069722e+04	1.038611e-12	1.029955e+16	2.442313e-13	2.590823e+19	400	9.998371e-01	1.804831e-02	71	85
47	1.052061e+04	1.024962e-12	1.026439e+16	2.525001e-13	2.660285e+19	400	9.998407e-01	1.784721e-02	71	85
48	1.035486e+04	1.011299e-12	1.023917e+16	2.606481e-13	2.732650e+19	400	9.998423e-01	1.775816e-02	71	85
49	1.019890e+04	9.978703e-13	1.022067e+16	2.323188e-13	2.426850e+19	400	9.998422e-01	1.776581e-02	71	85
50	1.002919e+04	9.827486e-13	1.020525e+16	2.402476e-13	2.502109e+19	400	9.998424e-01	1.775441e-02	71	85
51	9.870355e+03	9.682427e-13	1.019409e+16	2.480422e-13	2.577642e+19	400	9.998385e-01	1.796905e-02	71	85
52	9.721049e+03	9.543381e-13	1.018617e+16	2.557201e-13	2.653301e+19	400	9.998334e-01	1.825451e-02	71	85
53	9.580348e+03	9.410340e-13	1.018066e+16	2.276546e-13	2.359545e+19	400	9.998271e-01	1.859474e-02	71	85
54	9.427032e+03	9.263420e-13	1.017662e+16	2.351196e-13	2.434984e+19	400	9.998214e-01	1.890133e-02	71	85
55	9.283310e+03	9.124214e-13	1.017437e+16	2.424561e-13	2.509851e+19	400	9.998118e-01	1.939907e-02	72	85
56	9.148022e+03	8.991966e-13	1.017355e+16	2.496804e-13	2.584221e+19	400	9.998015e-01	1.992440e-02	72	85
57	9.020359e+03	8.866210e-13	1.017386e+16	2.567978e-13	2.658048e+19	400	9.997905e-01	2.046693e-02	72	85
58	8.899621e+03	8.746497e-13	1.017507e+16	2.281096e-13	2.361665e+19	400	9.997791e-01	2.101983e-02	72	85
59	8.767607e+03	8.614805e-13	1.017737e+16	2.350307e-13	2.434421e+19	400	9.997686e-01	2.151103e-02	72	85
60	8.643396e+03	8.490219e-13	1.018042e+16	2.418343e-13	2.506391e+19	400	9.997542e-01	2.216905e-02	72	85
61	8.526082e+03	8.371980e-13	1.018407e+16	2.485350e-13	2.577687e+19	400	9.997395e-01	2.282439e-02	72	85
62	8.415037e+03	8.259582e-13	1.018821e+16	2.551376e-13	2.648320e+19	400	9.997245e-01	2.347333e-02	72	85
63	8.309710e+03	8.152570e-13	1.019275e+16	2.616465e-13	2.718301e+19	400	9.997092e-01	2.411443e-02	72	85
64	8.209616e+03	8.050533e-13	1.019761e+16	2.317868e-13	2.410378e+19	400	9.996937e-01	2.474689e-02	72	85
65	8.099664e+03	7.938099e-13	1.020353e+16	2.381225e-13	2.479141e+19	400	9.996800e-01	2.529796e-02	72	85
66	7.995698e+03	7.831449e-13	1.020973e+16	2.443552e-13	2.547124e+19	399	9.900240e-01	1.408986e-01	72	85
67	7.897065e+03	7.729988e-13	1.021614e+16	2.504973e-13	2.614428e+19	399	9.898268e-01	1.422777e-01	72	85
68	7.803310e+03	7.633308e-13	1.022271e+16	2.565527e-13	2.681074e+19	399	9.896325e-01	1.436225e-01	72	85
69	7.714035e+03	7.541041e-13	1.022940e+16	2.625253e-13	2.747083e+19	399	9.894411e-01	1.449355e-01	72	85
70	7.628886e+03	7.452860e-13	1.023619e+16	2.320919e-13	2.431847e+19	399	9.892523e-01	1.462189e-01	72	85
71	7.535016e+03	7.355481e-13	1.024408e+16	2.379106e-13	2.496663e+19	399	9.891806e-01	1.467029e-01	72	85
72	7.445920e+03	7.262863e-13	1.025205e+16	2.436382e-13	2.560745e+19	399	9.889589e-01	1.481901e-01	72	85
73	7.361102e+03	7.174542e-13	1.026003e+16	2.492852e-13	2.624181e+19	399	9.887403e-01	1.496416e-01	72	85
74	7.280219e+03	7.090191e-13	1.026801e+16	2.548550e-13	2.686991e+19	399	9.885256e-01	1.510534e-01	72	85
75	7.202968e+03	7.009519e-13	1.027598e+16	2.603509e-13	2.749196e+19	399	9.883146e-01	1.524279e-01	72	85
76	7.129078e+03	6.932261e-13	1.028391e+16	2.657758e-13	2.810815e+19	399	9.881071e-01	1.537675e-01	72	85
77	7.058301e+03	6.858177e-13	1.029180e+16	2.711326e-13	2.871869e+19	399	9.879029e-01	1.550736e-01	72	85
78	6.990422e+03	6.787057e-13	1.029964e+16	2.390138e-13	2.535519e+19	399	9.877020e-01	1.563483e-01	72	85
79	6.915187e+03	6.708176e-13	1.030860e+16	2.442429e-13	2.595500e+19	399	9.876180e-01	1.568777e-01	72	85
80	6.843367e+03	6.632789e-13	1.031748e+16	2.493964e-13	2.654835e+19	399	9.873851e-01	1.583372e-01	72	85
81	6.774635e+03	6.560587e-13	1.032626e+16	2.544825e-13	2.713591e+19	399	9.871561e-01	1.597589e-01	72	85
82	6.708771e+03	6.491347e-13	1.033494e+16	2.595039e-13	2.771789e+19	399	9.869317e-01	1.611394e-01	72	85
83	6.645572e+03	6.424870e-13	1.034351e+16	2.644631e-13	2.829444e+19	399	9.867117e-01	1.624809e-01	72	85

Iter	$\sigma_{\max}$	$\sigma_{\min}$	κ	$\eta/\sigma^2_{\max}$	$\eta/\sigma^2_{\min}$	rank	$\|F_r\|/\|F\|$	$\|F_\perp\|/\|F\|$	$N_{<10^{-6}}$	$N_{<10^{-2}}$
84	6.584860e+03	6.360974e-13	1.035197e+16	2.693622e-13	2.886573e+19	399	9.864960e-01	1.637855e-01	72	85
85	6.526471e+03	6.299494e-13	1.036031e+16	2.742035e-13	2.943192e+19	399	9.862844e-01	1.650547e-01	72	85
86	6.470255e+03	6.240278e-13	1.036854e+16	2.789890e-13	2.999314e+19	399	9.860769e-01	1.662901e-01	72	85
87	6.416076e+03	6.183189e-13	1.037665e+16	2.453232e-13	2.641512e+19	399	9.858732e-01	1.674932e-01	72	85
88	6.355736e+03	6.119606e-13	1.038586e+16	2.500034e-13	2.696688e+19	399	9.857882e-01	1.679928e-01	72	85
89	6.297838e+03	6.058564e-13	1.039493e+16	2.546213e-13	2.751301e+19	399	9.855546e-01	1.693581e-01	72	85
90	6.242167e+03	5.999859e-13	1.040386e+16	2.591832e-13	2.805405e+19	399	9.853257e-01	1.706848e-01	72	85
91	6.188581e+03	5.943341e-13	1.041263e+16	2.636911e-13	2.859014e+19	399	9.851021e-01	1.719705e-01	72	85
92	6.136947e+03	5.888876e-13	1.042125e+16	2.681469e-13	2.912144e+19	399	9.848836e-01	1.732173e-01	72	85
93	6.087147e+03	5.836340e-13	1.042973e+16	2.725524e-13	2.964807e+19	399	9.846701e-01	1.744269e-01	72	85
94	6.039071e+03	5.785621e-13	1.043807e+16	2.769092e-13	3.017016e+19	399	9.844614e-01	1.756012e-01	72	85
95	5.992618e+03	5.736615e-13	1.044626e+16	2.812188e-13	3.068783e+19	399	9.842573e-01	1.767416e-01	72	85
96	5.947697e+03	5.689225e-13	1.045432e+16	2.854828e-13	3.120121e+19	399	9.840577e-01	1.778496e-01	72	85
97	5.904223e+03	5.643363e-13	1.046224e+16	2.504955e-13	2.741886e+19	399	9.838624e-01	1.789266e-01	72	85
98	5.855601e+03	5.592090e-13	1.047122e+16	2.546726e-13	2.792396e+19	399	9.837887e-01	1.793318e-01	72	85
99	5.808739e+03	5.542665e-13	1.048005e+16	2.587984e-13	2.842419e+19	399	9.835663e-01	1.805472e-01	72	85
100	5.763493e+03	5.494951e-13	1.048871e+16	2.628777e-13	2.891996e+19	399	9.833492e-01	1.817262e-01	72	85

Per-iteration residual energy diagnostics

Iter	$\sum \lvert U'F \rvert^2$	$E_{\sigma<10^{-6}}$	$E_{\sigma<10^{-4}}$	$E_{\sigma<10^{-2}}$	$E_{\sigma\geq 10^{-2}}$	weighted inv sens	$\sigma_{50\%}$	$\sigma_{90\%}$	$\sigma_{99\%}$
1	2.834590e+02	3.150565e-27	5.390466e-06	8.516775e-03	9.914832e-01	4.954243e+03	2.500514e+00	7.908430e-01	5.935443e-02
2	2.059762e+02	1.541234e-07	2.113946e-05	4.063652e-03	9.959363e-01	5.389123e+06	2.515024e+00	7.390953e-01	4.455327e-02
3	1.961631e+02	7.696183e-09	5.319523e-06	1.051427e-03	9.989486e-01	3.469045e+07	2.502904e+00	5.853437e-01	8.348641e-02
4	1.073207e+02	9.069972e-12	9.069972e-12	1.176897e-02	9.882310e-01	8.186349e+04	1.938203e+00	4.715087e-01	1.352938e-03
5	6.205985e+01	8.967985e-09	4.333527e-05	1.196943e-04	9.998803e-01	4.052667e+07	1.817121e+00	6.248568e-01	1.764825e-02
6	6.199370e+01	9.036080e-09	4.326143e-05	4.136325e-03	9.958637e-01	4.082666e+07	1.898230e+00	4.875301e-01	5.329919e-02
7	6.163252e+01	8.967951e-09	3.451231e-05	1.211765e-04	9.998788e-01	5.003730e+06	1.895429e+00	5.351229e-01	7.761311e-02
8	5.610575e+01	5.073865e-08	2.097989e-06	1.145258e-04	9.998855e-01	7.283835e+05	1.879509e+00	5.882108e-01	7.640237e-02
9	4.570602e+01	4.193871e-08	1.496870e-07	1.870742e-04	9.998129e-01	5.589034e+05	2.096727e+00	6.788376e-01	7.227464e-02
10	3.097880e+01	2.145836e-08	3.852666e-08	9.124058e-05	9.999088e-01	6.490177e+05	1.730169e+00	6.723803e-01	8.473372e-02
11	3.066996e+01	9.757635e-09	1.837835e-08	7.456154e-05	9.999254e-01	5.130079e+05	1.527112e+01	7.241150e-01	8.432245e-02
12	2.381218e+01	6.017825e-09	2.110656e-08	1.901794e-04	9.998098e-01	5.357051e+05	2.316496e+04	1.788888e+00	1.901597e-01
13	8.840471e+00	1.231691e-08	6.531421e-07	1.786867e-03	9.982131e-01	1.311664e+06	3.908207e+01	3.908207e+01	5.253708e+00
14	2.590302e-01	3.859082e-07	5.741390e-05	5.474651e-02	9.452535e-01	3.411182e+07	1.978397e+01	1.221819e+00	3.428759e-03
15	8.298488e-02	1.204316e-06	1.862499e-04	1.594346e-01	8.405654e-01	1.042380e+08	1.221059e+01	6.076265e-03	2.986821e-03
16	3.157989e-02	3.239400e-06	3.025292e-04	3.790712e-01	6.209288e-01	2.647552e+08	6.240348e+00	2.956256e-03	1.185770e-03
17	2.501259e-02	4.224952e-06	6.617626e-05	4.275537e-01	5.724463e-01	3.290650e+08	1.896181e-01	2.876753e-03	1.008198e-03
18	1.269869e-02	8.487443e-06	6.398853e-05	7.868884e-01	2.131116e-01	6.416044e+08	4.734540e-03	2.820159e-03	9.089490e-04
19	1.185384e-02	9.300264e-06	4.192105e-05	7.817590e-01	2.182410e-01	6.846643e+08	4.599927e-03	2.740200e-03	8.117777e-04
20	1.014230e-02	1.101633e-05	4.891718e-05	8.732460e-01	1.267540e-01	7.978188e+08	2.886640e-03	2.686792e-03	7.608159e-04
21	9.619912e-03	1.179129e-05	4.220907e-05	8.751249e-01	1.248751e-01	8.403464e+08	2.832880e-03	2.620867e-03	7.086445e-04
22	9.059704e-03	1.270376e-05	3.931819e-05	8.848085e-01	1.151915e-01	8.916893e+08	2.783659e-03	2.553799e-03	6.631105e-04
23	8.589039e-03	1.361760e-05	3.729546e-05	8.837056e-01	1.162944e-01	9.404184e+08	2.730718e-03	2.730718e-03	6.172008e-04
24	8.065799e-03	1.472460e-05	3.680128e-05	8.928390e-01	1.071610e-01	1.001407e+09	2.681263e-03	2.681263e-03	5.773154e-04
25	7.642448e-03	1.580246e-05	5.062902e-04	8.888158e-01	1.111842e-01	1.057523e+09	2.627279e-03	2.627279e-03	5.372473e-04
26	7.153583e-03	1.714940e-05	3.091946e-04	8.977141e-01	1.022859e-01	1.130578e+09	2.576422e-03	2.576422e-03	5.025281e-04
27	6.723306e-03	1.851734e-05	2.180888e-04	9.048914e-01	9.510862e-02	1.204166e+09	2.528307e-03	2.528307e-03	4.721390e-04
28	6.341075e-03	1.990623e-05	1.727980e-04	9.106675e-01	8.933251e-02	1.278414e+09	2.482717e-03	2.482717e-03	4.453309e-04
29	6.020294e-03	2.128176e-05	1.433429e-04	9.050438e-01	9.495621e-02	1.349158e+09	2.432854e-03	2.432854e-03	4.180295e-04
30	5.656015e-03	2.296854e-05	3.001501e-04	9.108927e-01	8.910726e-02	1.438835e+09	2.385981e-03	2.385981e-03	3.940397e-04
31	5.331266e-03	2.468399e-05	1.711485e-04	9.155317e-01	8.446830e-02	1.529751e+09	2.341877e-03	2.341877e-03	3.727748e-04
32	5.060333e-03	2.636791e-05	9.398433e-04	9.081475e-01	9.185245e-02	1.616237e+09	2.294015e-03	2.294015e-03	3.510146e-04
33	4.749482e-03	2.845515e-05	8.730871e-04	9.129689e-01	8.703113e-02	1.726790e+09	2.249330e-03	2.249330e-03	3.317973e-04
34	4.471943e-03	3.058091e-05	8.101784e-04	9.166789e-01	8.332110e-02	1.839293e+09	2.207535e-03	2.207535e-03	3.146839e-04
35	4.242230e-03	3.264635e-05	7.330218e-04	9.076330e-01	9.236695e-02	1.945961e+09	2.162426e-03	2.162426e-03	2.970911e-04
36	3.975903e-03	3.523991e-05	6.738556e-04	9.116131e-01	8.838690e-02	2.083585e+09	2.120470e-03	2.120470e-03	2.814808e-04
37	3.737982e-03	3.788610e-05	6.202545e-04	9.145638e-01	8.543623e-02	2.224107e+09	2.081314e-03	2.081314e-03	2.675195e-04
38	3.524227e-03	4.058322e-05	5.733979e-04	9.166187e-01	8.338126e-02	2.367493e+09	2.044629e-03	2.044629e-03	2.549554e-04
39	3.343492e-03	4.322959e-05	5.185361e-04	9.069002e-01	9.309980e-02	2.506212e+09	2.004853e-03	2.004853e-03	2.419060e-04
40	3.136730e-03	4.652680e-05	4.780895e-04	9.092956e-01	9.070440e-02	2.682272e+09	1.967433e-03	1.967433e-03	2.302095e-04
41	2.950915e-03	4.989872e-05	4.435067e-04	9.109360e-01	8.906404e-02	2.862613e+09	1.931381e-03	1.931381e-03	2.196535e-04
42	2.796644e-03	5.315072e-05	4.046567e-04	8.996158e-01	1.003842e-01	3.034688e+09	1.906578e-03	1.885995e-03	2.086583e-04
43	2.617169e-03	5.728859e-05	3.788320e-04	9.017484e-01	9.825164e-02	3.256929e+09	1.865627e-03	1.829990e-03	1.987751e-04
44	2.456210e-03	6.152877e-05	3.584539e-04	9.031734e-01	9.682660e-02	3.484992e+09	1.831290e-03	1.773823e-03	1.898330e-04
45	2.311249e-03	6.586663e-05	3.428755e-04	9.039583e-01	9.604174e-02	3.718606e+09	1.799513e-03	1.799513e-03	1.817005e-04
46	2.189983e-03	7.005301e-05	3.257414e-04	8.921986e-01	1.078014e-01	3.942620e+09	1.764939e-03	1.764939e-03	1.731661e-04
47	2.049460e-03	7.538680e-05	3.185229e-04	8.936223e-01	1.063777e-01	4.230615e+09	1.732250e-03	1.732250e-03	1.654392e-04
48	1.923085e-03	8.086182e-05	3.153523e-04	8.945366e-01	1.054634e-01	4.526457e+09	1.701129e-03	1.701129e-03	1.584031e-04
49	1.809037e-03	8.647070e-05	3.156240e-04	8.949721e-01	1.050279e-01	4.829703e+09	1.671360e-03	1.671360e-03	1.519674e-04
50	1.713855e-03	9.184490e-05	3.152192e-04	8.828206e-01	1.171794e-01	5.118964e+09	1.638405e-03	1.638405e-03	1.451758e-04
51	1.603113e-03	9.874817e-05	3.228868e-04	8.840613e-01	1.159387e-01	5.492586e+09	1.606877e-03	1.606877e-03	1.389943e-04
52	1.503421e-03	1.058396e-04	3.332272e-04	8.849232e-01	1.150768e-01	5.876505e+09	1.576636e-03	1.366375e-03	1.333391e-04
53	1.413398e-03	1.131081e-04	3.457642e-04	8.854128e-01	1.145872e-01	6.270102e+09	1.547578e-03	1.323365e-03	1.281450e-04
54	1.339120e-03	1.199630e-04	3.572603e-04	8.728542e-01	1.271458e-01	6.640169e+09	1.515355e-03	1.276516e-03	1.226419e-04
55	1.251660e-03	1.645676e-04	3.763238e-04	8.743167e-01	1.256833e-01	7.124741e+09	1.484559e-03	1.232399e-03	1.176143e-04
56	1.172937e-03	1.765853e-04	3.969817e-04	8.754789e-01	1.245211e-01	7.622756e+09	1.455119e-03	1.190808e-03	1.129997e-04
57	1.101885e-03	1.889383e-04	4.188953e-04	8.763211e-01	1.236789e-01	8.133318e+09	1.426987e-03	1.151535e-03	1.087491e-04
58	1.037474e-03	2.016277e-04	4.418333e-04	8.769261e-01	1.230739e-01	8.656474e+09	1.400122e-03	1.114403e-03	1.048208e-04
59	9.841479e-04	2.136396e-04	4.627246e-04	8.650295e-01	1.349705e-01	9.146270e+09	1.370572e-03	1.073958e-03	1.006340e-04
60	9.213107e-04	2.292867e-04	1.798368e-02	8.666704e-01	1.333296e-01	9.788605e+09	1.342606e-03	1.035914e-03	9.678645e-05
61	8.645814e-04	2.453950e-04	1.832835e-02	8.681052e-01	1.318948e-01	1.044834e+10	1.316142e-03	1.000124e-03	9.323668e-05
62	8.132481e-04	2.619344e-04	1.866849e-02	8.692952e-01	1.307048e-01	1.112428e+10	1.291109e-03	9.664244e-04	8.995145e-05
63	7.666148e-04	2.789018e-04	1.900458e-02	8.702980e-01	1.297020e-01	1.181641e+10	1.267435e-03	9.346679e-04	8.690231e-05
64	7.241011e-04	2.962931e-04	1.933699e-02	8.711574e-01	1.288426e-01	1.252466e+10	1.245048e-03	9.047201e-04	8.406476e-05
65	6.885585e-04	3.127263e-04	1.945742e-02	8.605399e-01	1.394601e-01	1.318778e+10	1.220660e-03	8.722141e-04	8.101971e-05
66	6.468008e-04	3.340281e-04	1.985242e-02	8.623837e-01	1.376163e-01	1.405340e+10	1.197795e-03	8.417580e-04	7.820225e-05
67	6.089169e-04	3.558918e-04	2.024293e-02	8.640811e-01	1.359189e-01	1.494091e+10	1.176336e-03	8.132315e-04	7.558657e-05
68	5.744813e-04	3.782738e-04	2.062742e-02	8.655857e-01	1.344143e-01	1.584873e+10	1.156182e-03	7.864976e-04	7.315182e-05
69	5.430739e-04	4.011633e-04	2.100631e-02	8.669338e-01	1.330662e-01	1.677665e+10	1.137237e-03	7.614329e-04	7.087989e-05
70	5.143396e-04	4.245485e-04	2.137996e-02	8.681535e-01	1.318465e-01	1.772443e+10	1.119411e-03	7.379277e-04	6.875500e-05
71	4.904727e-04	4.462691e-04	2.152175e-02	8.581114e-01	1.418886e-01	1.859943e+10	1.100078e-03	7.125948e-04	6.646334e-05
72	4.620847e-04	4.746873e-04	2.196030e-02	8.602640e-01	1.397360e-01	1.975218e+10	1.082024e-03	6.890698e-04	6.433211e-05
73	4.362471e-04	5.037385e-04	2.239260e-02	8.622701e-01	1.377299e-01	2.093135e+10	1.065127e-03	6.672857e-04	6.234415e-05
74	4.126904e-04	5.333590e-04	2.281712e-02	8.640833e-01	1.359167e-01	2.213471e+10	1.049289e-03	6.471809e-04	6.048546e-05
75	3.911466e-04	5.635267e-04	2.323428e-02	8.657319e-01	1.342681e-01	2.336179e+10	1.034419e-03	6.287193e-04	5.874379e-05
76	3.713873e-04	5.942187e-04	2.364443e-02	8.672387e-01	1.327613e-01	2.461207e+10	1.020436e-03	6.118699e-04	5.710838e-05
77	3.532171e-04	6.254097e-04	2.404781e-02	8.686205e-01	1.313795e-01	2.588498e+10	1.007264e-03	5.965578e-04	5.556975e-05
78	3.364648e-04	6.570794e-04	2.444479e-02	8.698973e-01	1.301027e-01	2.718010e+10	9.948372e-04	5.826521e-04	5.411949e-05
79	3.223839e-04	6.862668e-04	2.461060e-02	8.612703e-01	1.387297e-01	2.837559e+10	9.813105e-04	5.680709e-04	5.254309e-05
80	3.056416e-04	7.242087e-04	2.507068e-02	8.632980e-01	1.367020e-01	2.993604e+10	9.686284e-04	5.547739e-04	5.106522e-05
81	2.902884e-04	7.627151e-04	2.552289e-02	8.651979e-01	1.348021e-01	3.152502e+10	9.567026e-04	5.425063e-04	4.967629e-05
82	2.761888e-04	8.016994e-04	2.596590e-02	8.669329e-01	1.330671e-01	3.313968e+10	9.454661e-04	5.310798e-04	4.836843e-05

Iter	$\sum \lvert U'F \rvert^2$	$E_{\sigma<10^{-6}}$	$E_{\sigma<10^{-4}}$	$E_{\sigma<10^{-2}}$	$E_{\sigma\geq10^{-2}}$	weighted inv sens	$\sigma_{50\%}$	$\sigma_{90\%}$	$\sigma_{99\%}$
83	2.632067e-04	8.411235e-04	2.640006e-02	8.685231e-01	1.314769e-01	3.477919e+10	9.348590e-04	5.203630e-04	4.713465e-05
84	2.512241e-04	8.809492e-04	2.682568e-02	8.699857e-01	1.300143e-01	3.644271e+10	9.248276e-04	5.102620e-04	4.596875e-05
85	2.401386e-04	9.211381e-04	2.724305e-02	8.713357e-01	1.286643e-01	3.812941e+10	9.153239e-04	5.007068e-04	4.486519e-05
86	2.298609e-04	9.616520e-04	2.765240e-02	8.725859e-01	1.274141e-01	3.983846e+10	9.063048e-04	4.916431e-04	4.381902e-05
87	2.203126e-04	1.002453e-03	2.805398e-02	8.737478e-01	1.262522e-01	4.156904e+10	8.977316e-04	4.830266e-04	4.282581e-05
88	2.122382e-04	1.039319e-03	2.822159e-02	8.661766e-01	1.338234e-01	4.315670e+10	8.883210e-04	4.735774e-04	4.173824e-05
89	2.025708e-04	1.087345e-03	2.868217e-02	8.679501e-01	1.320499e-01	4.522031e+10	8.794224e-04	4.646281e-04	4.071092e-05
90	1.936355e-04	1.135630e-03	2.913329e-02	8.696245e-01	1.303755e-01	4.731090e+10	8.709820e-04	4.561372e-04	3.973855e-05
91	1.853668e-04	1.184062e-03	2.957385e-02	8.711723e-01	1.288277e-01	4.942506e+10	8.629616e-04	4.480688e-04	3.881675e-05
92	1.776981e-04	1.232587e-03	3.000422e-02	8.726096e-01	1.273904e-01	5.156170e+10	8.553272e-04	4.403914e-04	3.794160e-05
93	1.705708e-04	1.281149e-03	3.042476e-02	8.739506e-01	1.260494e-01	5.371977e+10	8.480482e-04	4.330768e-04	3.710954e-05
94	1.639335e-04	1.329695e-03	3.083578e-02	8.752078e-01	1.247922e-01	5.589824e+10	8.410972e-04	4.260994e-04	3.631742e-05
95	1.577409e-04	1.378171e-03	3.123759e-02	8.763924e-01	1.236076e-01	5.809614e+10	8.344495e-04	4.194363e-04	3.556234e-05
96	1.519528e-04	1.426525e-03	3.163048e-02	8.775140e-01	1.224860e-01	6.031251e+10	8.280829e-04	4.130666e-04	3.484170e-05
97	1.465335e-04	1.474707e-03	3.201473e-02	8.785816e-01	1.214184e-01	6.254643e+10	8.219771e-04	4.069713e-04	3.415315e-05
98	1.419526e-04	1.516490e-03	3.215989e-02	8.719304e-01	1.280696e-01	6.457075e+10	8.152123e-04	4.002468e-04	3.339421e-05
99	1.363880e-04	1.571927e-03	3.265523e-02	8.735496e-01	1.264504e-01	6.720892e+10	8.087546e-04	3.938404e-04	3.267245e-05
100	1.312027e-04	1.626967e-03	3.307087e-02	8.751146e-01	1.248854e-01	6.986880e+10	8.025729e-04	3.877299e-04	3.198494e-05